

Day 49 — TNPSC Group 2A Study Material

Part A – General Studies: Tamil Literature from the Sangam Age till Contemporary Times

READING MATERIAL

Covers the structure of the Sangam literary corpus (Tolkappiyam, the Ettuthogai and Pathuppattu anthologies, and the Twin Epics), plus a cluster of real exam facts about early Tamil-Brahmi inscriptions and 20th-century Tamil literary/political figures' newspaper associations.

Q: What is Tolkappiyam, and what are the Ettuthogai and Pathuppattu?

A: Tolkappiyam is the oldest surviving Tamil grammar text, foundational to the Sangam literary corpus. Ettuthogai ('Eight Anthologies') is a collection of 8 Sangam-era poetic works (including Purananooru, Kuruntogai, Akananooru, etc.). Pathuppattu ('Ten Idylls'/Pattuppattu) is a separate collection of 10 longer Sangam-era poems (100-800 lines each), including Nakkirar's well-known Tirumurukarrupadai.

Q: What are the 'Twin Epics' of Tamil literature, who wrote them, and what are they about?

A: Silappadhikaram and Manimekalai. Silappadhikaram, by Ilango Adigal, is the EARLIEST Tamil epic — the story of Kannagi and Kovalan, depicting Tamil culture, town planning, and the arts of dance/music. Manimekalai, by Sittalai Sattanar, is its (incomplete) sequel, following Kovalan and Madhavi's daughter, told from a Buddhist perspective.

Q: Real exam: What significant archaeological literary evidence has been found at Mangulam (Maankulam), near Madurai?

A: Tamil-Brahmi inscriptions (confirmed by a real exam question, which specifically noted 'five' such inscriptions found there) — among the earliest evidence of Tamil script, associated with Jain monks. The same exam question's other tested claims (about Jainism's specific entry route via the Andhra border, and an exact count of 'bed caves' near Madurai) were NOT confirmed correct — worth treating cautiously rather than as settled facts.

Q: Real exam: Which Tamil literary/political figures are correctly matched to their associated newspapers — Thiru.Vi.Ka, Subramania Bharathiyar, Annie Besant, Kasthuri Ranga Iyengar?

A: CORRECT pairings: Thiru.Vi.Ka (Thiru. Vi. Kalyanasundaranar) — 'Desabakthan'; Annie Besant — 'New India'. The exam's other two pairings were a deliberate SWAP trap: Subramania Bharathiyar was NOT associated with 'The Hindu' (his real papers were 'India' and 'Vijaya'); Kasthuri Ranga Iyengar was NOT associated with 'India' (he was actually associated with 'The Hindu') — the two were swapped with each other as wrong distractors.

MOCK TEST (WITH ANSWERS)

1. Which is the oldest surviving Tamil grammar text, foundational to the Sangam corpus?

- A. Thirukkural
- B. Tolkappiyam ✓**
- C. Silappadhikaram
- D. Naladiyar

Explanation: Tolkappiyam. [medium, web-research]

2. 'Ettuthogai' refers to a collection of how many Sangam-era anthologies?

- A. Five
- B. Eight ✓**
- C. Ten
- D. Eighteen

Explanation: Eight ('Eight Anthologies'). [medium, web-research]

3. 'Pathuppattu' ('Ten Idylls') is a separate Sangam-era collection of how many longer poems?

- A. Five
- B. Eight
- C. Ten ✓**
- D. Twelve

Explanation: Ten. [medium, web-research]

4. Which Tamil epic, by Ilango Adigal, is considered the EARLIEST Tamil epic, telling the story of Kannagi and Kovalan?

A. Manimekalai

B. Silappadhikaram ✓

C. Thirukkural

D. Naladiyar

Explanation: Silappadhikaram. [medium, web-research]

5. Manimekalai, the incomplete sequel to Silappadhikaram, was written by whom, and reflects which religious perspective?

A. Ilango Adigal; Jain perspective

B. Sittalai Sattanar; Buddhist perspective ✓

C. Thiruvalluvar; Hindu perspective

D. Avvaiyar; Jain perspective

Explanation: Sittalai Sattanar; Buddhist perspective. [hard, web-research]

6. Real exam: Tamil-Brahmi inscriptions have been found at which site near Madurai?

A. Keezhadi

B. Mangulam (Maankulam) ✓

C. Arikamedu

D. Adichanallur

Explanation: Mangulam (Maankulam). [medium, question-bank-2026]

7. Which newspaper was Thiru.Vi.Ka (Thiru. Vi. Kalyanasundaranar) associated with?

A. The Hindu

B. India

C. Desabakthan ✓

D. New India

Explanation: Desabakthan. [hard, question-bank-2024]

8. Subramania Bharathiyar was associated with which newspapers (NOT 'The Hindu', a common wrong pairing)?

A. India and Vijaya ✓

B. New India and Desabakthan

C. The Hindu and Swadesamitran

D. Young India and Harijan

Explanation: India and Vijaya. [hard, question-bank-2024]

9. Annie Besant was associated with which newspaper?

A. The Hindu

B. Desabakthan

C. New India ✓

D. India

Explanation: New India. [medium, question-bank-2024]

Part B – Aptitude & Mental Ability: Simplification

READING MATERIAL

Simplification is 1 of 10 named Aptitude sub-skills (15 questions total for the whole Aptitude unit). TNPSC's simplification questions are rarely 'plug into one formula' — they combine BODMAS, fractions/decimals, ratios, and algebraic shortcuts. Each item below is a worked method, not just an answer — the goal is to recognize which technique a given expression calls for.

Q: Method — BODMAS with a nested bracket and an unknown: Solve for x : $57 - [42 - \{37 - (20 - x)\}] = 176 \div (8 \div 2)$

A: Work from the innermost bracket outward, and simplify the RHS first: $176 \div (8 \div 2) = 176 \div 4 = 44$. Now the LHS: $(20 - x)$ stays as is for now $\rightarrow \{37 - (20 - x)\} = 17 + x \rightarrow [42 - (17 + x)] = 25 - x \rightarrow 57 - (25 - x) = 32 + x$. Set equal to 44: $32 + x = 44 \rightarrow x = 12$. (Real TNPSC-style problem; this exact expression is a recurring coaching-site reference question.)

Q: Method — sum of two squares without brute-force squaring: Find $217 \times 217 + 183 \times 183$.

A: Use $a^2 + b^2 = (a+b)^2 - 2ab$ instead of squaring two 3-digit numbers by hand. Here $a+b = 400$ (round number — that's the clue this trick applies) and $ab = 217 \times 183 = 39,711$. So $a^2+b^2 = 400^2 - 2(39,711) = 160,000 - 79,422 = 80,578$. Recognizing 'the two numbers add to a round number' is the trigger for this shortcut.

Q: Method — difference of squares shortcut: Simplify $(197 \times 197 - 103 \times 103) \div (197 - 103)$ without long multiplication.

A: $a^2 - b^2 = (a+b)(a-b)$, so $(a^2-b^2)/(a-b) = (a+b)$ directly — no multiplication needed at all. Here $a=197$, $b=103$, so the answer is simply $197+103 = 300$. This 'difference of squares divided by the difference' shape is a favorite TNPSC trick question precisely because it looks like it needs heavy calculation but collapses to one addition.

Q: Method — fraction-of-decimal addition: Simplify $7/5$ of $58 + 3/8$ of 139.2 .

A: $7/5$ of $58 = 7 \times 58/5 = 406/5 = 81.2$. $3/8$ of $139.2 = 3 \times 139.2/8 = 417.6/8 = 52.2$. Sum: $81.2 + 52.2 = 133.4$.

Q: Method — ratio substitution to simplify a compound ratio expression: If $x : y = 3 : 4$, find $(4x + 5y) : (5x - 2y)$.

A: Don't solve for x and y separately — substitute $x=3k$, $y=4k$ (k is just a scaling factor that will cancel). $(4x+5y) = 12k+20k = 32k$. $(5x-2y) = 15k-8k = 7k$. Ratio = $32k : 7k = 32 : 7$. The k always cancels in a ratio question — that's the whole trick.

Q: Method — same ratio-substitution trick, different ratio: If $x : y = 3 : 5$, find $(5x + 3y) : (15x - 2y)$.

A: Substitute $x=3k$, $y=5k$. $(5x+3y) = 15k+15k = 30k$. $(15x-2y) = 45k-10k = 35k$. Ratio = $30k:35k = 6:7$.

Q: Method — simultaneous equations feeding into a simplification target: If $3a + 4b = 22$ and $8a - 5b = -4$, find $(a+b)^2$.

A: Solve the pair normally: from $3a+4b=22$ and $8a-5b=-4$, elimination gives $a=2$, $b=4$ (check: $3(2)+4(4)=6+16=22$ ✓; $8(2)-5(4)=16-20=-4$ ✓). The question doesn't ask for a and b individually — it asks for $(a+b)^2 = (2+4)^2 = 36$. TNPSC often hides the real target $((a+b)^2, \text{ not } a \text{ or } b)$ at the end of a simultaneous-equation question — always re-read what's actually being asked before solving for individual variables.

Q: Method — mixed numbers with a nested fraction: Simplify $2\frac{1}{2} + 3\frac{1}{3} \div \frac{2}{3}$.

A: BODMAS still applies to fractions — division before addition. $3\frac{1}{3} \div \frac{2}{3} = (10/3) \div (2/3) = (10/3) \times (3/2) = 5$. Then $2\frac{1}{2} + 5 = 7\frac{1}{2}$. The common mistake is adding $2\frac{1}{2}+3\frac{1}{3}$ first — division always binds tighter than addition, fractions or not.

MOCK TEST (WITH ANSWERS)

1. Solve for x : $70 - [45 - \{33 - (15 - x)\}] = 171 \div (9 \div 3)$

- A. 10
- B. 12
- C. 14 ✓**
- D. 16

Explanation: RHS: $171 \div (9 \div 3) = 171 \div 3 = 57$. LHS inner-out: $\{33 - (15 - x)\} = 33 - 15 + x = 18 + x \rightarrow [45 - (18 + x)] = 27 - x \rightarrow 70 - (27 - x) = 43 + x$. Set $43 + x = 57 \rightarrow x = 14$. [medium, authored]

2. Find $312 \times 312 + 288 \times 288$ using the $(a+b)^2 - 2ab$ shortcut.

- A. 179,808
- B. 180,288 ✓**
- C. 190,000
- D. 182,688

Explanation: $a+b=600$, $ab=312 \times 288=89,856$. $a^2+b^2=600^2-2(89,856)=360,000-179,712=180,288$. [medium, authored]

3. Simplify $(425 \times 425 - 375 \times 375) \div (425 - 375)$.

- A. 400
- B. 750
- C. 800 ✓**
- D. 425

Explanation: Using $a^2 - b^2 = (a+b)(a-b)$, dividing by $(a-b)$ leaves just $(a+b) = 425+375 = 800$. [easy, authored]

4. Simplify $\frac{5}{6}$ of 72 + $\frac{2}{7}$ of 98.

- A. 88 ✓**
- B. 84
- C. 90
- D. 86

Explanation: $\frac{5}{6}$ of 72 = 60. $\frac{2}{7}$ of 98 = 28. Sum = 88. [easy, authored]

5. If $p : q = 5 : 6$, find $(3p + 2q) : (4p - q)$.

- A. 27 : 14 ✓**
- B. 9 : 7
- C. 27 : 7
- D. 14 : 9

Explanation: $p=5k, q=6k$. $(3p+2q)=15k+12k=27k$. $(4p-q)=20k-6k=14k$. Ratio = 27:14. [medium, authored]

6. If $m : n = 4 : 7$, find $(2m + 3n) : (5m - n)$.

- A. 29 : 13 ✓**
- B. 13 : 29
- C. 29 : 20
- D. 20 : 29

Explanation: $m=4k, n=7k$. $(2m+3n)=8k+21k=29k$. $(5m-n)=20k-7k=13k$. Ratio = 29:13. [medium, authored]

7. If $5a + 2b = 16$ and $3a - 4b = -6$, find the value of $(a+b)^2$.

- A. 16
- B. 25 ✓**
- C. 36
- D. 9

Explanation: Multiply the first equation by 2: $10a+4b=32$. Add to the second $(3a-4b=-6)$: $13a=26 \rightarrow a=2$. Substitute back: $5(2)+2b=16 \rightarrow b=3$. $(a+b)^2=(2+3)^2=25$. [hard, authored]

8. Simplify $4\frac{2}{3} + 2\frac{1}{4} \div \frac{3}{4}$.

- A. $7\frac{2}{3}$ ✓**
- B. $6\frac{1}{3}$
- C. 7
- D. $6\frac{2}{3}$

Explanation: $2\frac{1}{4} \div \frac{3}{4} = (9/4) \div (3/4) = 3$. Then $4\frac{2}{3} + 3 = 7\frac{2}{3}$. [medium, authored]

9. A trader borrows ₹16,000 at 12% p.a. simple interest and another ₹12,000 at 18% p.a. simple interest. In how many years will the two amounts owed (principal + interest) become equal?

- A. $12\frac{2}{3}$ years
- B. $14\frac{2}{3}$ years
- C. $16\frac{2}{3}$ years ✓**
- D. $18\frac{2}{3}$ years

Explanation: This mirrors a real 2025 TNPSC Aptitude question style (same numbers, Simple Interest unit) — included here as a reminder that 'Simplification' skills (careful fraction/ratio handling) carry directly into other Aptitude sub-topics like Simple Interest. Set $16000+16000(0.12)t = 12000+12000(0.18)t$ and solve for t to get $16\frac{2}{3}$ years. [hard, question-bank-2025]

10. Find the value of: $89 \times 89 + 11 \times 11 + 2 \times 89 \times 11$.

- A. 10,000 ✓**
- B. 9,801
- C. 8,900
- D. 10,201

Explanation: This is the $(a+b)^2$ pattern in disguise: $a^2+2ab+b^2 = (a+b)^2$. Here $a=89$, $b=11$, so the answer is $(89+11)^2 = 100^2 = 10,000$ — no separate multiplication needed. [medium, authored]

Part C – General English: Prose: "Sports Stars" — Mithali Dorai Raj

READING MATERIAL

A profile of Mithali Raj, India's most celebrated women's cricketer, who captained the national team in Tests and ODIs, known as the 'Tendulkar of Indian Women's Cricket' for her record-breaking consistency.

Q: Who is the central figure of this lesson, what sport does she play, and what formats did she captain India in?

A: Mithali Raj. She plays cricket, and captained the Indian women's team in both Test and ODI formats.

Q: What is Mithali Raj's mother tongue, where does she live, and at what age did she start playing cricket?

A: Her mother tongue is TAMIL. She lives in Hyderabad. She started playing cricket at age 10, and was selected for the Indian team at age 17.

Q: What is Mithali's nickname, and real exam: which award was she ACTUALLY honoured with (not Padma Bhushan, a false claim tested in a real exam)?

A: Nicknamed the 'Tendulkar of Indian Women's Cricket'. She was actually honoured with the PADMA SHRI (2015) — a real exam tested 'Padma Bhushan' as a deliberately false claim about her award.

Q: Real exam: Was Mithali discouraged from playing cricket by her family? What is the real story?

A: NO — a real exam flagged 'Mithali was not encouraged to play cricket by her family' as FALSE. In reality, her father, Dorai Raj (a retired Indian Air Force officer), actively introduced her to the sport by taking her along to her brother's cricket coaching sessions at a young age.

Q: Real exam: What specific consecutive-innings record is Mithali the first woman to achieve — and what's the common trap?

A: She is the first woman to score SEVEN CONSECUTIVE FIFTIES (half-centuries) — NOT seven consecutive centuries, which a real exam tested as a false claim. Mixing up 'fifties' with 'centuries' is the trap to watch for.

Q: What notable innings did Mithali play against South Africa in February 2018, and who else scored well in that match?

A: She scored 76 not out. Smriti Mandhana also hit a stylish half-century in the same match.

Q: Approximately how many runs has Mithali scored in Women's ODIs?

A: Over 6,000 runs.

MOCK TEST (WITH ANSWERS)

1. Who is the central sports figure discussed in this lesson?

- A. Smriti Mandhana
- B. Jhulan Goswami
- C. Mithali Raj ✓**
- D. Harmanpreet Kaur

Explanation: Mithali Raj. [easy, web-research]

2. Mithali Raj's mother tongue is:

- A. Hindi
- B. Tamil ✓**
- C. Telugu
- D. Kannada

Explanation: Tamil. [medium, web-research]

3. Mithali Raj's nickname is:

- A. Lady Dhoni
- B. Tendulkar of Indian Women's Cricket ✓**
- C. Queen of Bat
- D. Cricket Star

Explanation: Tendulkar of Indian Women's Cricket. [medium, web-research]

4. Real exam: Mithali Raj was honoured with which award (NOT Padma Bhushan, a false claim tested in a real exam)?

- A. Bharat Ratna
- B. Padma Shri ✓**

- C. Padma Bhushan
- D. Padma Vibhushan

Explanation: Padma Shri (2015). [hard, question-bank-2025]

5. Real exam: Was Mithali discouraged from playing cricket by her family?

- A. Yes, her family opposed it initially
- B. No — her father actively introduced her to the sport ✓**
- C. Her mother alone supported her
- D. She started without any family involvement

Explanation: No, her father actively introduced her to cricket. [hard, question-bank-2025]

6. Real exam: Mithali is the first woman to score seven consecutive:

- A. Centuries
- B. Fifties (half-centuries) ✓**
- C. Sixes in an over
- D. Man-of-the-match awards

Explanation: Fifties (half-centuries), not centuries. [hard, question-bank-2025]

7. At what age did Mithali start playing cricket, and at what age was she selected for the Indian team?

- A. Started at 7, selected at 15
- B. Started at 10, selected at 17 ✓**
- C. Started at 12, selected at 19
- D. Started at 14, selected at 20

Explanation: Started at 10, selected at 17. [medium, web-research]

8. What was Mithali's score in the February 2018 match against South Africa?

- A. 56*
- B. 66
- C. 76* ✓**
- D. 86

Explanation: 76 not out. [hard, web-research]

9. Who else scored a stylish half-century alongside Mithali in that same match against South Africa?

- A. Jhulan Goswami
- B. Harmanpreet Kaur
- C. Smriti Mandhana ✓**
- D. Shafali Verma

Explanation: Smriti Mandhana. [medium, web-research]

10. Approximately how many runs has Mithali Raj scored in Women's ODIs?

- A. Over 3,000
- B. Over 4,000
- C. Over 5,000
- D. Over 6,000 ✓**

Explanation: Over 6,000. [medium, web-research]